PREETHAM S SHINDE

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PROFESSIONAL SUMMARY

Computer Science student with expertise in AI-based attendance systems, full-stack development, and data structures. Proven track record of building scalable applications using Python, React, and modern technologies. Strong problem-solving skills demonstrated through multiple technical projects. Passionate about automation, algorithms, and creating practical solutions for real-world problems.

EDUCATION

Sri Chaitanya PU College | Pre-University (1st and 2nd Year)

Sri Lakshmi School | 10th Standard

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, Java, C

Frontend Technologies: React, HTML, CSS, JavaScript, Responsive Web Design

Backend Technologies: Flask, Python, Node.js, Database Design

AI/ML & Computer Vision: Face Recognition, Machine Learning Fundamentals, OpenCV, TensorFlow

Algorithms & Techniques: Dijkstra's Algorithm, Graph Algorithms, Kruskal's Algorithm, Prim's Algorithm, Union-Find, Recommendation Systems, Distributed Computing

Databases & Tools: MySQL, Supabase, Git, GitHub, VS Code, Plotly, Pandas, RFID, QR Code

Core CS Competencies: Data Structures, Algorithms, Graph Theory, Full Stack Development, Web Development

PROJECTS

Absentia – AI-Based Attendance Management System

Tech Stack: React, Python, Flask, Face Recognition, OpenCV, MySQL/Supabase, RFID/QR Code

• Engineered an AI-powered automated attendance management system to eliminate manual attendance marking and reduce administrative overhead

• Implemented face recognition using OpenCV and computer vision algorithms to accurately identify and verify student presence in real-time

• Integrated alternative identification methods (QR Code, RFID) for multi-modal student recognition and system reliability

• Developed Flask backend with Python for session management and attendance data processing with timestamps and subject tracking

• Built React-based teacher dashboard displaying real-time attendance status, present/absent counts, and automatic attendance percentage calculations

• Created student notification system alerting users when attendance falls below threshold percentages to improve engagement

• Designed automated report generation module for monthly and subject-wise attendance analytics and compliance tracking

• Key Achievement: Reduced attendance marking time from 10+ minutes per class to automated process in seconds

AI-Based Traffic Junction Optimization System

Tech Stack: React, Python, Dijkstra's Algorithm, Traffic Dataset, Data Analysis

• Developed intelligent traffic management system for mixed-traffic conditions using Python for traffic pattern analysis

• Implemented Dijkstra's algorithm for optimal route path calculation and dynamic congestion management

• Engineered dynamic signal priority management based on real-time traffic density analysis and historical patterns

• Integrated emergency vehicle detection and priority routing mechanisms for faster emergency response

• Created interactive React-based visualization dashboard for real-time traffic monitoring and signal optimization display

Minimum Spanning Tree Algorithm Visualizer

Tech Stack: React, JavaScript, Kruskal's Algorithm, Prim's Algorithm, 3D Visualization

• Built interactive 3D visualization tool for Minimum Spanning Tree algorithms enabling hands-on graph algorithm learning

• Implemented Kruskal's and Prim's algorithms with Union-Find data structure for efficient cycle detection

• Designed step-by-step algorithm execution visualization showing real-time edge selection and tree construction

• Created responsive UI with interactive graph manipulation enabling users to understand graph theory concepts intuitively

Inventory and Supply Chain Analytics System

Tech Stack: Python, Flask, Machine Learning, Plotly, Pandas, Data Analysis

• Developed full-stack web application for inventory management and supply chain analytics using Flask backend

• Implemented machine learning models for predictive inventory analysis and demand forecasting capabilities

• Built interactive Plotly heatmaps and dashboards visualizing inventory patterns, trends, and supply chain inefficiencies

• Integrated Pandas for ETL operations and data processing pipeline for automated inventory data transformation

Spotify Distributed Recommendation System

Tech Stack: Python, Machine Learning, Collaborative Filtering, Distributed Computing

• Developed music recommendation engine using collaborative filtering and machine learning techniques

• Implemented distributed computing concepts for scalable recommendation processing across large datasets

• Engineered user-song interaction analysis pipeline for generating accurate personalized recommendations

POP Works Professional Website

Tech Stack: React, Modern Web Development, GitHub, Responsive Design

• Developing responsive professional website using modern React architecture and component-based design

• Building interactive UI components with focus on user experience, accessibility, and performance optimization

• Implementing version control with Git and GitHub for collaborative development and project management

CORE COMPUTER SCIENCE COMPETENCIES

Data Structures: Arrays, Linked Lists, Trees, Graphs, Hash Tables, Stacks, Queues, Heaps

Algorithms: Sorting, Searching, Graph Traversal (BFS/DFS), Shortest Path, Dynamic Programming, Greedy Algorithms

Database Management: SQL, Database Design, Normalization, Query Optimization

Operating Systems: Process Management, Memory Management, Scheduling, Synchronization

Computer Networks: OSI Model, TCP/IP Protocol, Network Security Basics, Socket Programming

ACHIEVEMENTS

• Developed multiple real-world applications combining AI/ML, web development, and data structures

• Strong foundation in competitive programming and algorithm optimization

• Demonstrated expertise in full-stack development with Python backend and React frontend

• Continuous learner focused on AI/ML, automation, and emerging technologies

• Committed to creating practical solutions that solve real-world problems efficiently

LANGUAGES

English (Professional), Kannada (Native), Hindi (Conversational)